Unit – IV

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# Multipath Mitigation Techniques

## EQUALISATION:-

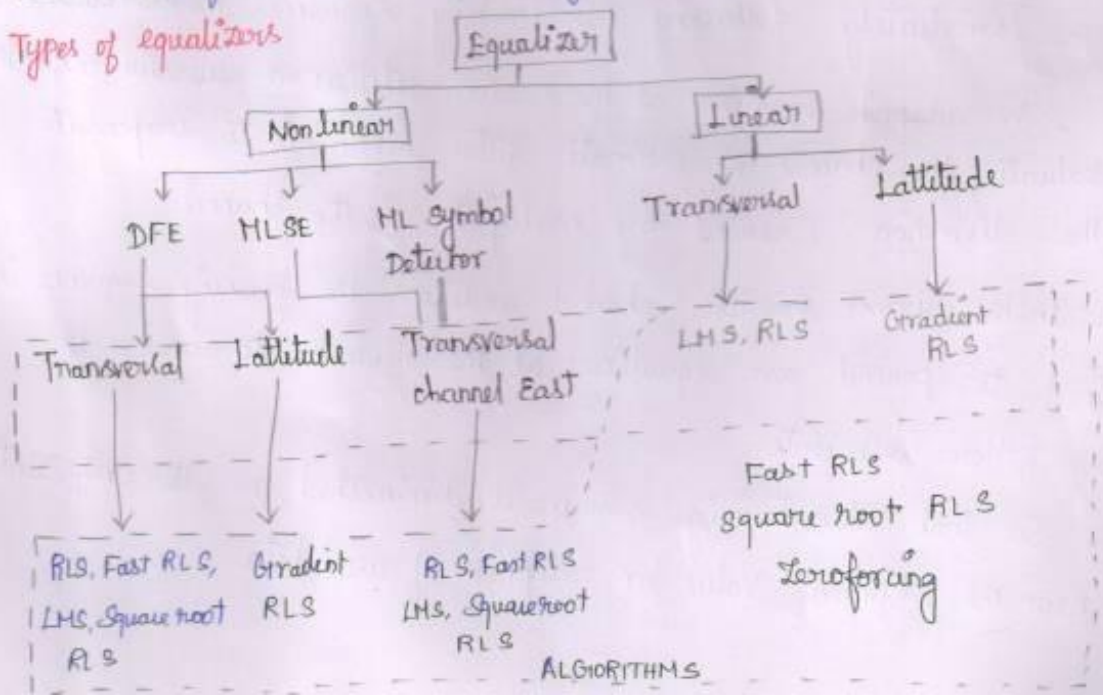
Three techniques which can be used independently or in tandem to improve received signal quality & link performance over small scale times & distances.

The techniques are

1) Equalization
2) Diversity
3) channel coding

Equalization can be used for any signal processing operation that minimizes ISI. In radio channels a variety of adaptive equalizers can be used to carried interference while providing diversity.

Types of equalizers



Adaptive equalizers:-

- Equalizer are used to track the time varying characteristic of the mobile channel & it is called adaptive equalizers since the mobile fading channel is random & time varying modes of an adaptive equalizer

The operation modes of an adaptive equalizer is

(i) training (ii) tracking

- First a fixed length training sequence is sent by the transmitter then the receivers equalizer may adapt to a proper setting for minimum bit error rate detection.

- The training sequence is a pseudorandom binary signal or a fixed prescribed bit pattern

- Immediately following this training sequences the user data is sent

- The adaptive equalizer at the receiver utilizes a recursive algorithm to evaluate the channel & estimate filter coefficients to compensate for the distortion created by multipath in the channel.

- In the worst possible channel conditions the training sequence is designed to permit an equalizer at the receiver to acquire the proper filter coefficients

- So that when training sequence is diminished the filter coefficients are near the optimal value for reception of user data

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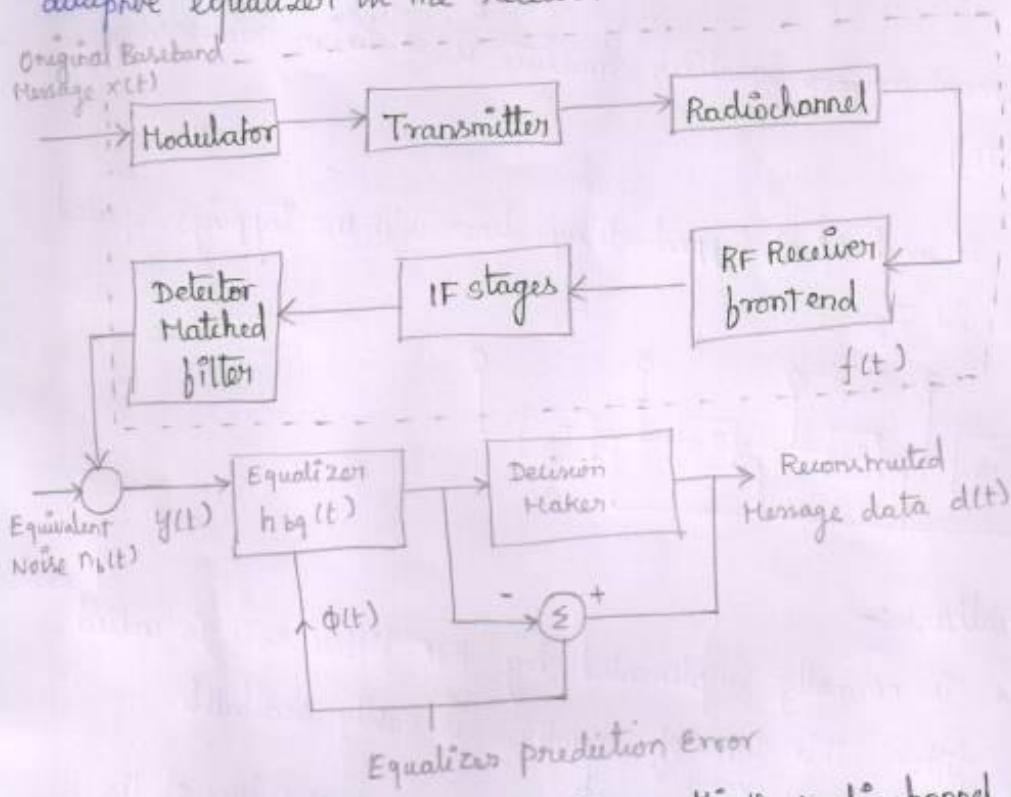
- As user data are received the adaptive algorithm of the equalizer tracks the changing channel.

- The block diagram of a communication system with an adaptive equalizer in the receiver




f(t) = combined IR of transmitter multipath radio channel & receiver RF/IF

The equalization techniques can be broadly classified into 2 types based upon the o/p of an adaptive equalizer.

1) Linear Equalizer
2) Non linear Equalizer

Linear Equalization :-

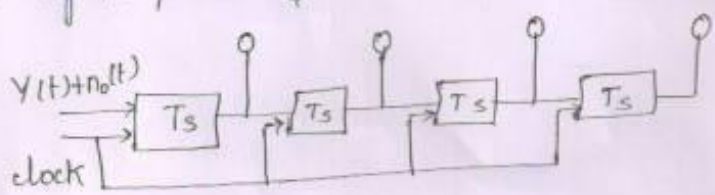
If the output d(t) is not used in the feedback path to adapt the equalizer then the type equalization is called linear equalization

Non-linear equalization :-

If d(t) is feedback to change the subsequent o/ps of the equalization then the type of equalization is called non-linear equalization

- The most common equalizer structure is a linear transverse equalizer (LTE).

A LTE is made up of tapped delay lines with the tappings spaced a symbol period $T_s$

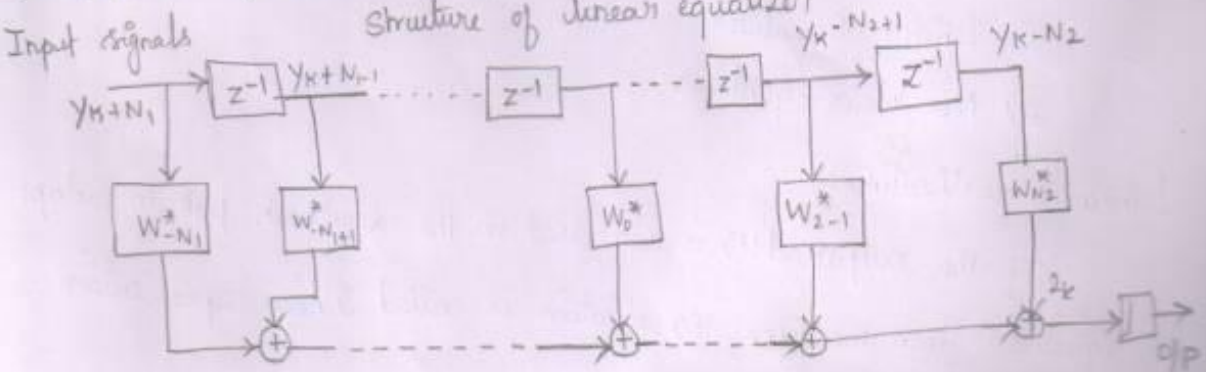



Linear Equalizers :-

It is normally implemented by FIR filter so it is called transversal filter. This type of equalizer is easily available

In this type of equalizer the present & past values of the received signals are linearly weighted by the filter coefficients & summed to produce the o/p $d_k$

Structure of linear equalizer



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From the o/p of this linear equalizer before a threshold direction is

$$\hat{d}_K = \sum_{n=-N_1}^{N_2} (W_n^*) y_{K-n}$$

Where

$W_n^*$ – Complex filter coefficients (or) tap weights

$\hat{d}_K$ – o/p at time index K

$y_i$ – The input received signal at time $t_0 + iT$

$t_0$ – Equalizer starting time

$N = N_1 + N_2 + 1$ is no. of taps

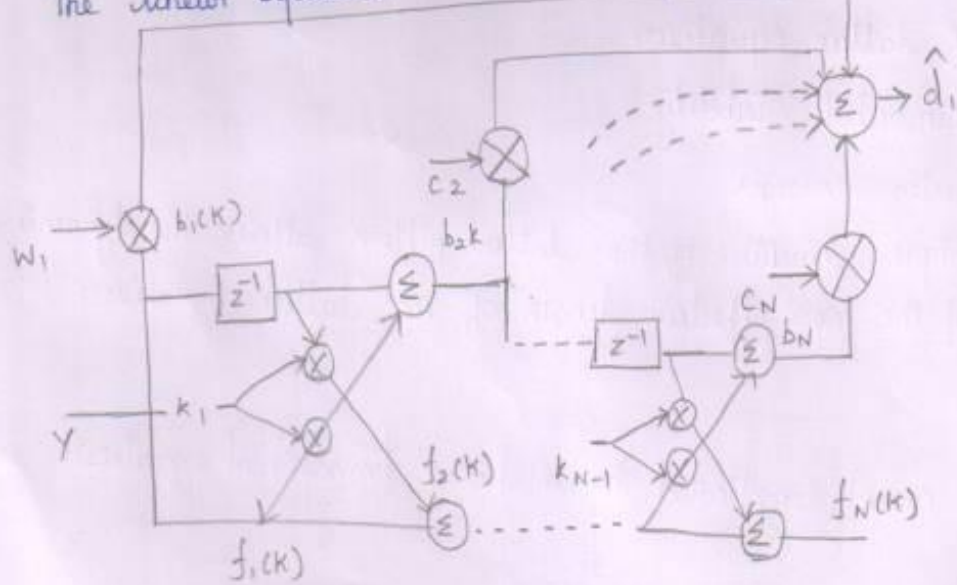
The minimum mean squared error $E(|e(n)^2|)$ that a linear equalizer can achieve is

$$E[|e(n)|^2] = \frac{T}{2\pi} \int_{-\pi/T}^{\pi/T} \frac{N_0}{|x(e^{j\omega T})|^2 + N_0} d\omega$$

$x(e^{j\omega T})$ – frequency response of the channel

$N_0$ – Noise Power spectral density

The linear equalizer can also be implemented by lattice filter



In this type of filter the i/p signal $y_k$ is transformed into set of n intermediate forward & backward error signals $f_n(k)$ & $b_n(k)$ respectively

The each stage of the lattice is then characterized by the following recursive equation

$$f_1(k) = b_1(k) = y(k)$$

$$f_n(k) = y(k) - \sum_{i=1}^{n} k_i\, y(k-i) = f_{n-1}(k) + k_{n-1}(k)\, b_{n-1}(k-1)$$

$$b_n(k) = y(k-n) - \sum_{i=1}^{n} k_i\, y(k-n+i) + b_{n-1}(k-1) + k_{n-1}(k)\, f_{n-1}(k)$$

$k_n(k)$ - reflection coefficient for the $n^{th}$ stage of the lattice

The $b_n$ backward error signals are than used as i/ps to the tap weights & the o/p of the equalizer is given by

$$\hat{d}_k = \sum_{n=1}^{N} W_n(k)\, b_n(k)$$

Advantage of lattice equalizer

- Numerical stability
- Faster coverages
- Unique structure of the lattice filters allows the dynamic assignment of the most effective length of the lattice equalizer.

Disadvantage

It is more complicated than a linear transversal equalizer

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## Non- linear Equalization :-

If the channel distortion is too serve in linear equalizer to handle then the non-linear equalizers are used to compensate the distortion

There are 3 non-linear methods are used in most 2G & 3G systems.

These are

1) Decision Feedback Equalization
2) Maximum likelihood Symbol Detection
3) Maximum likelihood Sequence Estimation

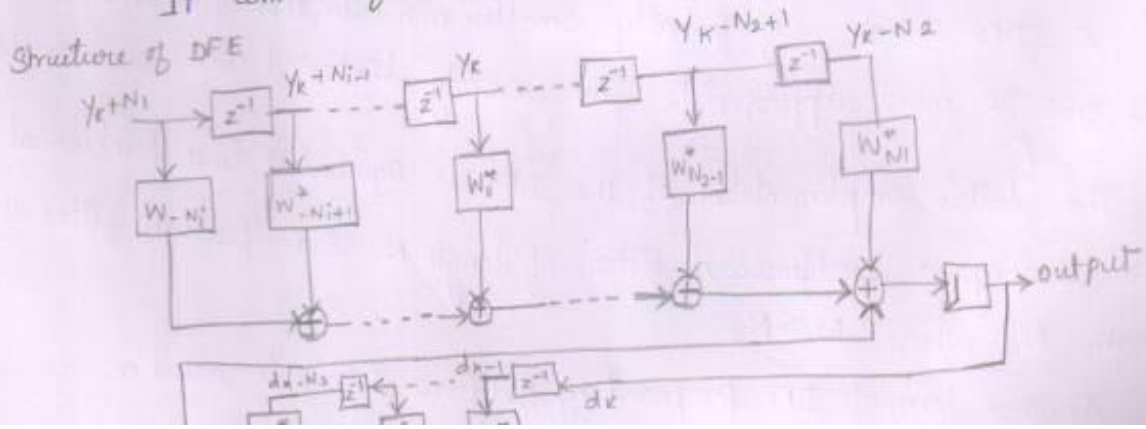
## Decision Feedback equalization

The basic idea used in the DFE is if the further symbols induce ISI that can be estimated and subtracted out before detection of subsequent symbols.

The DFE can be realized by direct transversal form or as a lattice filter

It consist of a feed forward filter & a feedback filter

Structure of DFE



The FBF is driven by decisions on the o/p of the detector & its coefficients can be adjusted to cancel the ISI on the current symbol from past detected symbols.

The equlizer has $N_1+N_2+1$ taps the feedforward filter & $N_3$ taps in the FBF & its o/p can be expressed as

$$\hat{d}_k = \sum_{n=-N_1}^{N_2} W_n^* y_{k-n} + \sum_{i=1}^{N_3} F_i d_{k-i}$$

$W_n^*$ & $y_n$ → Tap gains & the i/p to the feed filter

$F_i^*$ – Tap gains for the FBF

$d_i (i<k)$ – previous decision made on the detected signal

Once $\hat{d}_k$ is obtained than $d_k$ is decide from it. Then $d_{k-1}$, $d_{k-2}$ are feedback into the equalizer & $\hat{d}_{k+1}$ is obtained by using the above equation $d_k$.

The minimum mean squared error a DFE can achieve is

$$E(|e(n)|^2)_{min} = \exp\left\{\frac{T}{2\pi}\int_{-\pi/T}^{\pi/T} \ln\left[\frac{N_0}{|X(e^{j\omega T})|^2+N_0}\right]\right.$$

A DFE has significantly smaller minimum MSE than an LTE. The DFE is more appropriate for severely distorted wireless channels

The lattice implementation of the DFE is equivalent to a transversal DFE having a feedforward filter of length $N_1$ & a feedback filter of length $N_2$ where $N_1 > N_2$

Another form of DFE proposed by Belfiore & park is called a

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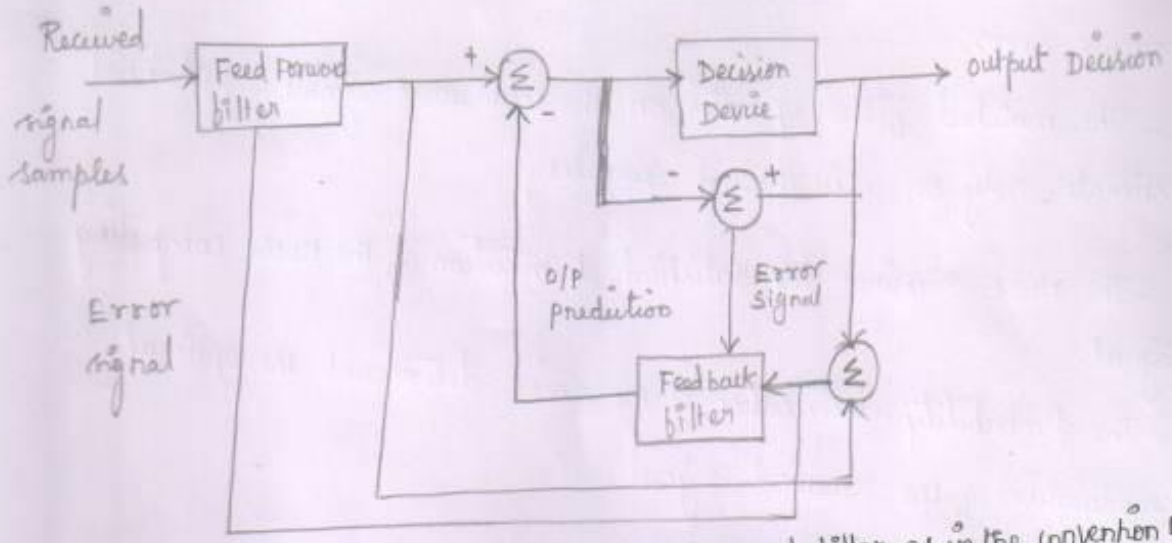
Predictive Decision feedback equalizer



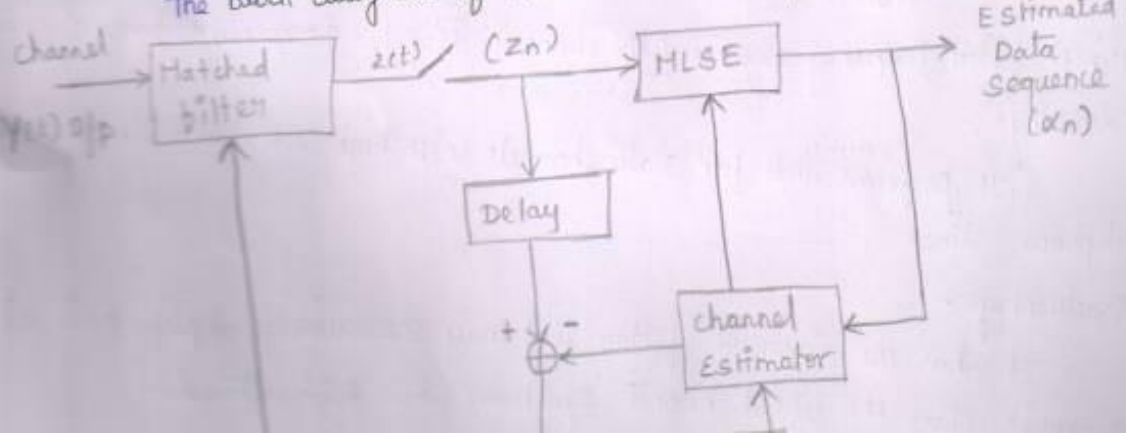

- The equalizer consists of a feedforward filter as in the convention DFE. The FBF is called a Noise predictor because it predicts the Noise & the residue ISI contained in the signal at the FFF O/P & subtracts from it the detector O/P after some feedback delay.

Maximum likelihood Sequence Estimation (MLSE) Equalizer

- In this type of equalizer we are using a channel impulse response Simulator within the algorithm.

- The MLSE Test all possible data sequences & chooses the data sequence with the maximum probability as the O/P.

The block diagram of MLSE with an adaptive matched filter



- The MLSE is optimal in the sense that it minimizes the probability of a sequence error.
- The matched filter operates on the continuous signal & channel estimate rely on discretized samples.
- The MLSE requires the satistical distribution of the noise computing signal.
- The Probability distribution of the noise determines the optimal demodulation of the received signal.

Zero forcing algorithm [ZFA]

- Zero forcing means set equalizer impulse response to zero & the equalizer coefficients $W_n$ are chosen to force the samples of the combined channel.
- In this algorithm the delay elements provide a time delay equal to the symbol duration T & the frequency response $H_{eq}(f)$ of the equalizer is periodic with a period combined equal to the symbol rate 1/T.

The combined response of the channel with the equalizer is

$$H_{ch}(f)\, H_{eq}(f) = 1, \quad |f| < 1/2T$$

$H_{ch}(f)$ - folded frequency response of the channel $H_{eq}(f)$ - frequency response of equalizer

Advantage:-

- It performs well for static channels with high SNR such as local wired telephone lines

Disadvantage:-

It has the inversive filter that may excessively amplify noise at frequencies where the folded channel spectrum has high attenuation.

Least Mean Square algorithm (LMSA) (6)

In order to minimize the mean square error b/w the desired equalizer o/p & the actual equalizer o/p the LMS equalizer is used.

The predictor error is given by

$$e_K = d_K - \hat{d}_K = x_K - \hat{d}_K$$

One more equation for $e_K$ is

$$e_K = x_K - Y_K^T W_K = x_K - W_K^T Y_K$$

The mean square error $|e_K|^2$ at time instant K is obtained by

$$\mathcal{E} = E(e_K^* e_K)$$

For a specific channel condition the prediction error $e_K$ is dependent on the tap gain vector $W_N$ so that mean square error of an equalizer is a function of $W_N$.

Let the cost function $J(W_N)$ denote the mean squared error as a function of tap gain vector $W_N$

In order to minimize the MSE

$$\frac{d}{dW_N} J(W_N) = -2P_N + 2R_{NN} W_N = 0$$

$$R_{NN} \hat{W}_N = P_N$$

Then the MMSE of an equalizer is

$$J_{opt} = J(\hat{W}_N) = E(X_K X_K^*) - P_N^T \hat{W}_N$$

$$\hat{W} = R_{NN}^{-1} P_N$$

The LMSA is the simplest equalization algorithm and requires only 2N+1 operation per iteration

Advantages:

1. Low computation of complexity
2. Simple program

Disadvantage

1. Poor tracking
2. Slow convergence.

Diversity Techniques

It is a powerful technique which is used in the receiver to improve the efficiency of the wireless link with relatively low cost.

- In this type of reception the received signal is selected from many paths among those the strongest signal is predicted

- By placing many antennas in different directions, diversity reception can be achieved. This is termed as antenna diversity or space diversity.

For diversity reception technique

$$P(\gamma_1, \gamma_2 \ldots \gamma_m \leq \gamma) = (1 - e^{-\gamma/\Gamma})^M = P_M(\gamma)$$

M - no. of paths / diversity branch

$\Gamma$ - Mean SNR

$\gamma_1, \gamma_2 \ldots \gamma_M$ - SNR of each branch

$\gamma$ - Threshold SNR

$P(\gamma_1, \gamma_2 \ldots \gamma_M)$ - Probability of SNRs falling below a threshold $\gamma$.

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Micro diversity :-

The diversity methods that combat Small-Scale fading i.e the fading created by interference of MPCs is called as Micro diversity

TYPES

* Temporal diversity
* spatial diversity
* Frequency diversity
* Angular diversity
* polarization diversity

TEMPORAL DIVERSITY :-

As the wireless propagation channel is time Variant signals that are received at different times are uncorrelated. For sufficient de correlation

The Temporal distance must be atleast $1/2 f_{max}$

$f_{max}$ - Maximum Doppler frequency.

Temporal diversity can be realized by following ways

1. Repetition coding
2. Automatic ~~Request~~ Repeat Request
3. Combination of interleaving & coding

Repetition coding

The signal is repeated several times where the repetition intervals are long enough to achieve de-correlation

To achieves diversity but is also highly b.w inefficient. spetral efficiency ↓ by a factor that is equal to the number of repetitions.

Disadvantage:- Retransmission Occur always

Automatic Repeat Request:

The Rx sends a message to the Tx to ensure whether it received the data with sufficient quality. If the data is not transmitted successfully then the transmission is repeated

Advantage :-

1. The spectral efficiency of ARQ is better than that of repetition coding

2. It requires multiple transmission only when the 1st transmission Occur in a bad fading state.

Disadvantage:- It requires feedback channel

Combination of interleaving & coding

A more advanced version of repetition coding is forward error correction coding with interleaving. The different symbols of a code word are transmitted at different times which ↑ the probability that at least some of them arrive with a good SNR.

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## Space Diversity (or) Antenna Diversity

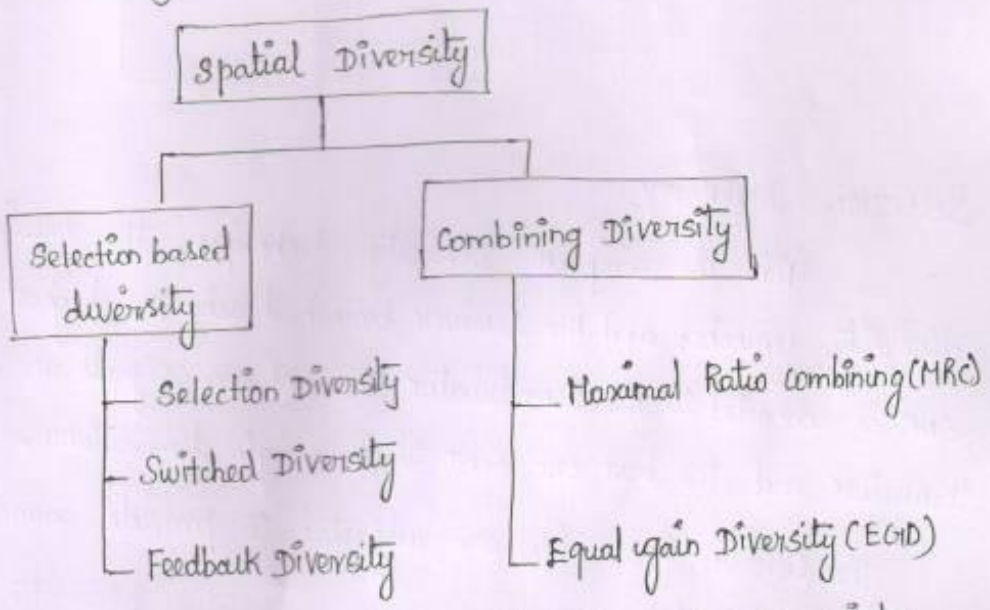
### Principle

The basic principle of this type of diversity is selecting the best signal among all the signals received from different branches at the receiving end.

Space diversity reception methods can be classified as

a) Selection Based Diversity

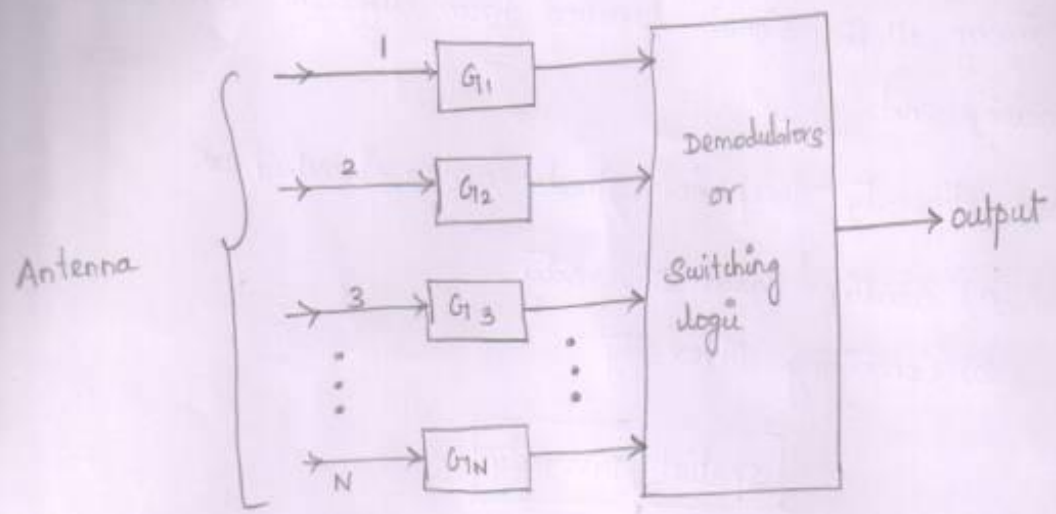
b) Combining Diversity




- One of the most popular forms of diversity used in the wireless system is space diversity also known as antenna diversity.
- This diversity technique is used in base station design
- At each cell site, multiple base station receiving antennas are used to provide diversity reception

- The base station antennas are spaced considerably far apart to achieve decorrelation

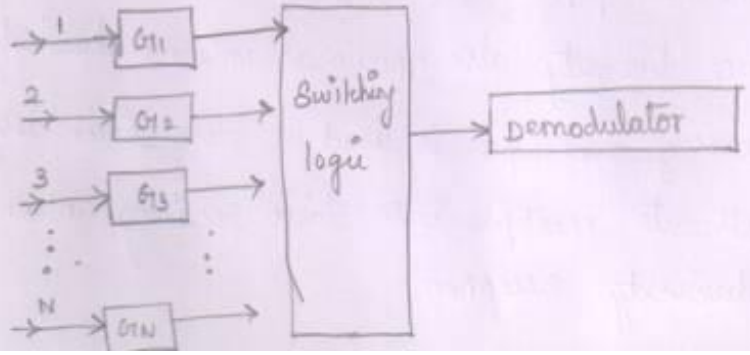
Block diagrams for space diversity




SELECTION DIVERSITY:

This the simplest Diversity technique. The receiver has M diversity branches and the receiver branch having the highest instaneous SNR is connected to the demodulator. The antenna signals are sampled themselves and the best one sent to a single demodulator.

The $G_1, G_2, G_3 \ldots G_N$ are adjusted to provide same average SNR to all branches Block diagram of selection Diversity



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Switched Diversity

Principle

Selection Criterion of active diversity branch is monitored. If it falls below a certain thershold, then the Rx switches to a different antenna. Switching only depends on the quality of the active diversity branch: It does not matter whether the other branch actually provides a better signal quality or not.

Needs:-

Selection diversity has selection criteria that all diversity branches have to be monitored in order to know when to select a different antenna. This leads to either ↑ hardware effort or reduced spectral efficiency. To overcome this selection diversity drawbacks switched diversity is used.

Parameters in Switched diversity

1. Switching thershold 2. Hysteresis time

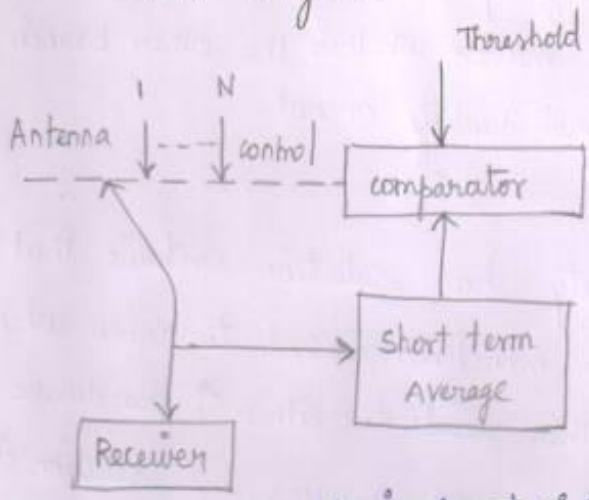
FEEDBACK OR SCANNING DIVERSITY

Principle

Scanning diversity is similar to selection diversity except that instead of always using the best of M signals. The M signals are scanned in a fixed sequence until one is found to be above a predetermined thershold. This signal is received until it falls below thershold & the scanning process is again initiated

- Instead of selecting best among N signals the N i/ps are scanned one by one until an i/p above a thershold SNR is found.
- This i/p is used until the SNR falls below the thershold & the scanning is again initiated

Block diagram




Advantage: It is very simple to implement & only one receiver is required. Disadvantage: The resulting fading statistics are somewhat inferior to those obtained by other methods.

Maximal Ratio Combining:-

In this method the signals from all of the N branches are weighted according to their individual signal voltage to noise power ratios and then summed.

- An individual receiver and phasing circuit for each antenna element & the individual signals must be co-phased before being summed.
- It produces an o/p SNR equal to the sum of individual SNR's

Block diagram

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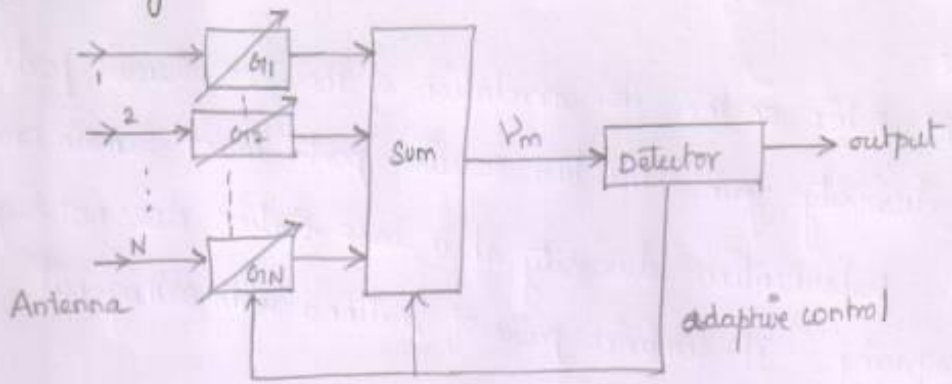



Advantage:-

It gives the best satistical reduction of fading of any known linear diversity combines

Disadvantage:-

1. It requires individual receiver & phasing circuits for each antenna elements.
2. Needs high implementation cost.

Equal gain diversity

- In this method the branch weights are all set to unity but the signals from each branch are cophased to provide equal gain combining diversity
- This allows the receiver to exploit the signals that are simultaneously received on each branch.

Advantage:- Performance is only margenally inferior to maximal ratio combining & superior to selection diversity.

Polarization Diversity:-

Principle

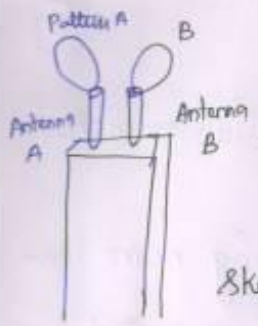
It relies on the de-correlation of the two receive ports to achieve diversity gain. The two receiver ports must remain cross polarized. polarization diversity at a base station does not require antenna spacing. It combines pairs of antenna with orthogonal polarizations

- In this diversity the transmitted signal with horizontal or vertical polarization is received by antenna with two elements.
- One element is used for horizontal polarization & the other is used for vertical polarization. Wireless communication systems usually use vertical polarization because this is more convinent for use with portable & mobile antennas.
- A vertically polarized signal may be transformed into horizontal polarization due to multipath propagation the signal received in any polarization will be intercepted by the 2 element antennas.

The 2 elements are polarized $\pm\alpha$ to the vertical assuming vertical polarization employed in the transmitter

Advantage:

It reduces the multipath delay spread.




The mutual coupling which depicts that antenna B acts as reflector for antenna A. whose pattern is skewed to left. By the by Antenna B pattern is skewed to right due to reflections from antenna A.

TIME Diversity

- The signal representing the same information are sent over the same channel at different times. Time diversity repeately transmits information at time spacings that exceeds the coherence time of the channel. Multiple repetition of the signal will be received with independent fading conditions thereby providing for diversity.

- The information is transmitted at time spacings that exceeds the coherence time of the channel.

Disadvantage:- It requires spare B.W also as many receivers as there are channels used for the frequency diversity

One modern implementation of time diversity involves the use of the RAKE receiver for spread spectrum CDMA

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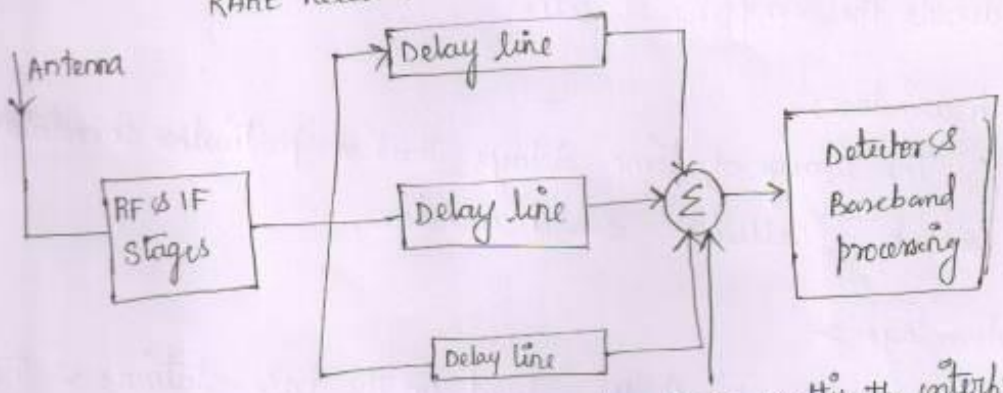
## RAKE RECEIVER:-

- The CDMA receivers may combine the time delayed versions of the original signal transmission in order to improve the SNR at the receiver. This is done by RAKE receiver.

- It attempt to collect the time shifted versions of the original signal by providing a separate correlation receiver for each of the multipath signals. It provides link improvement through time diversity

RAKE Receiver




One of the advantage of CDMA system is the multipath interference can be reduced by combining direct & reflected signal in the receiver. The receiver used are called RAKE receivers

The mobile unit can combine three Rf signals delaying 2 of them to match the 3rd. One of these signals can be assumed to be the base station in the current cell.

The other 2 may be reflection is neighbouring base stations.

Normally the base station receiver can combine 4 signals. The signals are
1. The direct signal from the mobile
2. Three reflected signal

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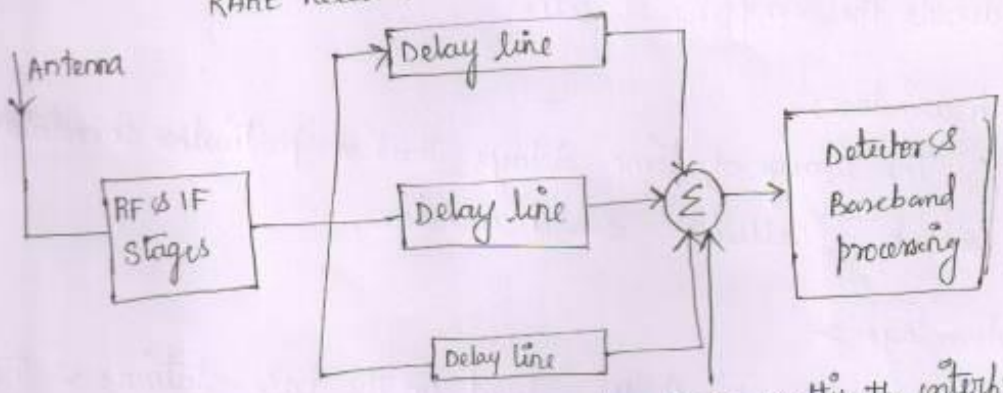
RAKE Receiver

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- In addition two base station's may receive a signal from the same mobile.
- Then the BS send their to the MSC which uses the higher quality signal. Decision about quality are made on a frame by frame basis every 20ms.

It is possible to have 2 base stations communicating with the mobile indefinitely in what is referred to as soft handoff. This avoids the dropping of calls.

Search window :-

The range of time delays that a particular correlator can search is called a search window.

Disadvantage :-

It considerably ↑ the load on the base stations & the switching network.

Macrodiversity

- The diversity methods the combat large scale fading i.e the fading created by shadowing effects is called as Macro diversity
- Shadowing is almost independent of transmit frequency & polarization so that frequency diversity or polarization diversity are not effective

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If correlation distance for long scale fading are on the order of tens or hundreds of meters, then the spatial diversity can be used.

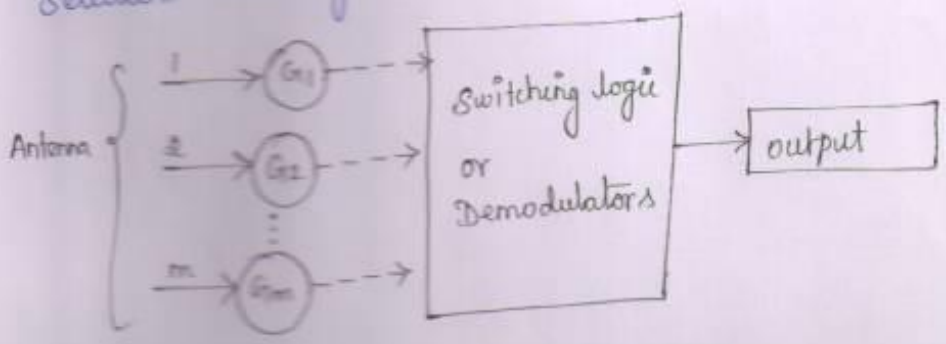
The simplest method for macrodiversity is the use of on frequency repeaters that receive the signal & retransmit an amplified version of it.

Advantage

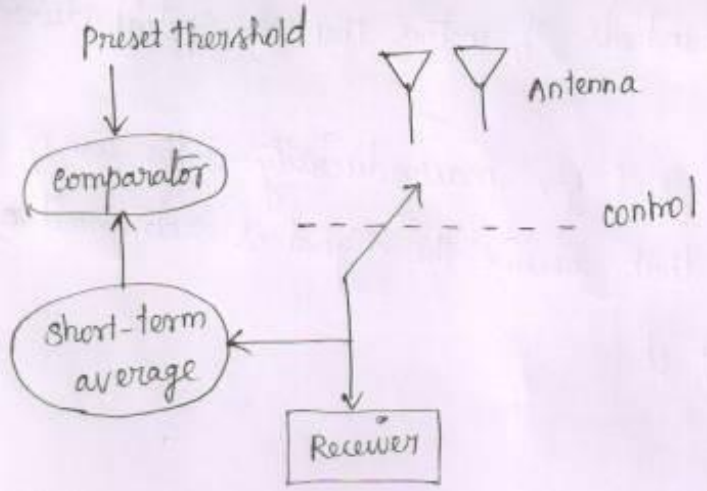
1. Use of on-frequency repeaters that receive signal & retransmit an amplified version of it
2. In cellular appliations the two BS should be synchronized
3. Compensating large signal fading Macro diversity is used.

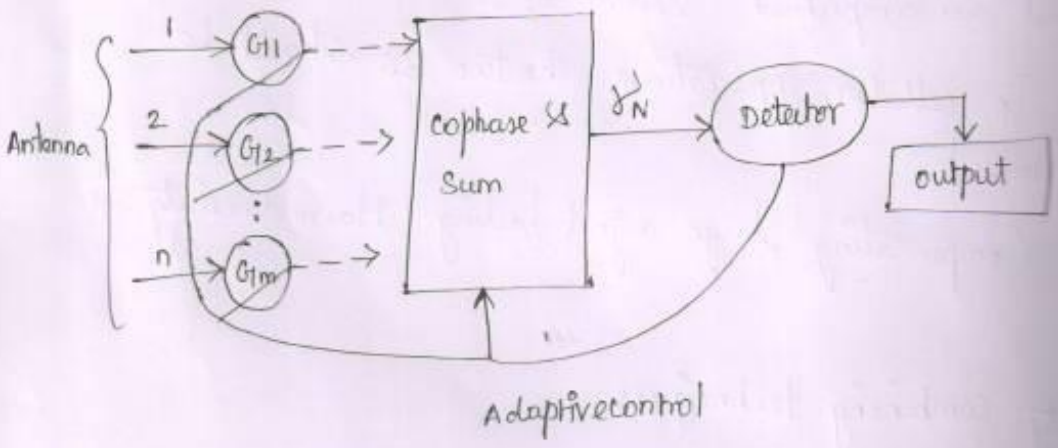
Diversity Combining techniques

Selection diversity



Feedback or Scanning diversity




Maximal ratio combining